YUBIN KIM

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Summary

Hello, I'm a final-year Ph.D. candidate at Personal Robots Group, MIT Media Lab and currently a Student Researcher at Google Research. I develop **Language Model-driven Agent Systems for Healthcare** with a focus on **Safety and Scalability**. My work spans multi-agent systems (NeurIPS 2024 Oral, top 0.39%), agent scaling laws, and AI safety mechanisms. I have 10+ publications at top venues (Science Robotics, NeurIPS, ICML, EMNLP, ACL, CVPR, etc) with multiple Nature-tier papers under review.

Education

Massachusetts Institute of Technology *Jan 2025 – Dec 2026 (expected)*
Ph.D. in Artificial Intelligence and Healthcare

Massachusetts Institute of Technology *Sep 2023 – Dec 2024*
M.S. in Media Arts and Sciences

Yonsei University *Mar 2014 – Feb 2022*
B.S. in Computer Science and Mechanical Engineering
2-year military service (Republic of Korea Marine Corps, 2016–2017)

Selected Publications

MDAgents: An Adaptive Collaboration of LLMs for Medical Decision-Making
Y. Kim, C. Park, H. Jeong, et al.
NeurIPS 2024 Oral (Top 0.39%) | arXiv | Code
Achieved 5.8% improvement over GPT-4 on medical benchmarks through adaptive multi-agent collaboration

Health-LLM: Large Language Models for Health Prediction via Wearable Sensor Data
Y. Kim, X. Xu, D. McDuff, C. Breazeal, H. Park
CHIL 2024 | arXiv | Project
First comprehensive framework for health prediction using LLMs with multimodal wearable data

Towards a Science of Scaling Agent Systems
Y. Kim, K. Gu, C. Park, et al.
Under Review at Nature Machine Intelligence | arXiv | Code | Blog
Established quantitative scaling laws across 180 configurations ($R^2 = 0.513$); discovered multi-agent coordination trade-offs

Medical Hallucination in Foundation Models and Their Impact on Healthcare
Y. Kim, H. Jeong, S. Chen, et al.
Under Review at Nature Medicine | Project
Comprehensive analysis of hallucination patterns across 12 foundation models on clinical tasks

The Anatomy of a Personal Health Agent
A. Heydari, ..., Y. Kim, et al.
Under Review at Nature | Blog
Comprehensive framework for personal health agents leveraging wearable sensor data

InvThink: Towards AI Safety via Inverse Reasoning
Y. Kim, T. Kim, E. Park, et al.
Under Review at ICML 2026 | Project
Novel safety mechanism using inverse reasoning to detect and prevent harmful AI outputs

Tiered Agentic Oversight: A Hierarchical Multi-Agent System for AI Safety in Healthcare
Y. Kim, H. Jeong, E. Park, et al.
Under Review at CHIL 2026 | Project
Hierarchical multi-agent oversight system ensuring AI safety in clinical decision-making

Industry Experience

Google Research *Jun 2025 – Present*
Research Intern | Supervisor: Dr. Daniel McDuff
- Developing agentic AI systems for novel biomarker discovery in wearables
- Building multi-agent frameworks for automated health data analysis at scale

Honda Research Institute USA *May 2024 – Aug 2024*
Research Intern | Supervisor: Dr. Nawid Jamali
- Developed LLM-enhanced human-human-robot interaction framework
- Designed social mediator agent behaviors validated through user studies

Bear Robotics *Mar 2019 – Mar 2022*
Software/Robotics Engineer Intern
- Developed Graph-SLAM algorithms for food service delivery robots (Servi, Pennybot)
- Improved localization robustness through multi-sensor fusion (LiDAR, IMU, RGB-D)

Technical Skills

Languages	Python, C++, Shell
ML/DL	PyTorch
Agent Systems	LangChain, LiteLLM
Infrastructure	SLURM, Docker, GCP
Robotics	ROS, MoveIt, SLAM, Gazebo, Unity

Full Publication List

Under Review / In Preparation

- Y. Kim et al., "CoDaS: AI Co-Data-Scientist for Novel Biomarker Discovery in Wearables" *Nature Medicine*
- Y. Kim et al., "Sensor-Agents: Universal Time-Series Sensor Simulator via LLM Agents" *Nature*
- Y. Kim et al., "Towards a Science of Scaling Agent Systems" [arXiv] [Blog] *Nat. Mach. Intell.*
- Y. Kim et al., "Medical Hallucination in Foundation Models" [Web] *Nature Medicine*
- A. Heydari, ..., Y. Kim et al., "The Anatomy of a Personal Health Agent" [Blog] *Nature*
- Y. Kim et al., "InvThink: Towards AI Safety via Inverse Reasoning" [Web] *ICML '26*
- Y. Kim et al., "Tiered Agentic Oversight: Hierarchical Multi-Agent System" [Web] *CHIL '26*

Published

- Y. Kim et al., "BehaviorSFT: Behavioral Token Conditioning for Clinical Agents" [Web] EMNLP '25 Findings
- Y. Kim et al., "Tiered Agentic Oversight for AI Safety in Healthcare" ICML MAS '25
- Y. Kim et al., "Vocal Agent: LLMs for Vocal Health Diagnostics" [arXiv] Interspeech '25 **Oral**
- H. Chen, Y. Kim et al., "Social Robots as Conversational Catalysts" [Paper] Science Robotics
- Y. Kim et al., "MDAgents: Adaptive Collaboration of LLMs for Medical Decision-Making" [arXiv] NeurIPS '24 **Oral**

- J. Shen*, Y. Kim* et al., “EmpathicStories++: Multimodal Dataset for Empathy” [arXiv] ACL ’24 Findings
- H. Ryu, ..., Y. Kim et al., “Diffusion-EDFs: Bi-equivariant Denoising on SE(3)” [arXiv] [Web] CVPR ’24
- Y. Kim et al., “Health-LLM: LLMs for Health Prediction via Wearable Sensor Data” [arXiv] [Web] CHIL ’24
- Y. Kim et al., “HIINT: Historical, Intra- and Inter-personal Dynamics Modeling” [arXiv] ICMI ’23
- D. Lee, Y. Kim et al., “Multipar-T: Multiparty-Transformer for Contingent Behaviors” [arXiv] IJCAI ’23
- Y. Kim et al., “Joint Engagement Classification using Video Augmentation” [arXiv] AAMAS ’22 **Oral**

Awards & Scholarships

Kwanjeong Scholarship (MS/PhD funding, $200K+) 2023 – 2027

Samsung × MIT Hackathon, 2nd Place 2022

Global Industrial Talent Scholarship 2022 – 2023

High Honor (Top 3%), Yonsei University 2019

References

Prof. Cynthia Breazeal
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